Single Image Dehazing

using a Model-Agnostic CNN

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Introduction

What is Image Dehazing?

Single image dehazing recovers a clear image from a single hazy input caused by fog, smoke, or atmospheric scattering.

Traditional Approach

Relies on the atmospheric scattering model:

$$I(x) = J(x) \cdot t(x) + A \cdot (1 - t(x))$$

Requires estimation of transmission map t(x) and atmospheric light A — making the problem ill-posed.

Our Approach

A model-agnostic CNN that directly learns: $J(x) = f(I(x))$
No intermediate parameter estimation required.

Input: Hazy Image

Target: Clear Image

Motivation

Autonomous Driving

Haze impairs obstacle detection and lane recognition, posing safety risks in self-driving systems.

Surveillance & Security

Outdoor cameras in foggy conditions produce poor-quality footage, hindering threat detection.

Remote Sensing

Satellite and aerial images are often degraded by atmospheric haze, affecting geographic analysis.

Deep Learning Advantage

End-to-end networks bypass hand-crafted priors, learning richer representations directly from data.

Related Work

Traditional Prior-Based Methods

Dark Channel Prior (DCP)

He et al., 2011

How it works:

Assumes at least one color channel is near-zero intensity in most non-sky patches. Uses this prior to estimate the transmission map t(x) and atmospheric light A.

PSNR: 18.87 dB · SSIM: 0.794

⚠ Drawbacks

- Fails on bright or white objects (sky, snow, white walls)
- Computationally expensive soft-matting step
- Assumes a single global atmospheric light — inaccurate in real scenes
- Cannot handle spatially varying illumination well

Color Attenuation Prior (CAP)

Zhu et al., 2015

How it works:

Models scene depth using the difference between brightness and saturation in HSV space. A linear model is learned to estimate depth from this difference.

Good on uniform haze; limited on dense fog

⚠ Drawbacks

- Strong assumption: depth proportional to brightness–saturation gap
- Fails when colors are desaturated for non-haze reasons
- Poor performance on non-homogeneous haze
- Sensitive to parameter tuning

Non-Local Image Dehazing (NLD)

Berman et al., 2016

How it works:

Exploits the observation that colors of a haze-free image can be well approximated by a few hundred distinct colors (haze-lines in RGB space).

Better on outdoor scenes with complex textures

⚠ Drawbacks

- Struggles in indoor or low-light environments
- Computationally expensive cluster analysis
- Requires careful selection of haze-line clusters
- Sensitive to noise and image compression artifacts

Related Work

Learning-Based Approaches & Common Drawbacks

DehazeNet

Cai et al., 2016

PSNR 22.66 dB · SSIM 0.833

Approach:
CNN that estimates the transmission map t(x), then plugs it into the atmospheric scattering model to recover J(x). Uses BReLU activation.

⚠ Limitation: Requires ground-truth t(x) maps for supervision; still model-dependent.

MSCNN

Ren et al., 2016

PSNR 20.01 dB · SSIM 0.791

Approach:
Multi-scale CNN with a coarse-scale network for global transmission and a fine-scale network for local refinement. Plug-in approach.

⚠ Limitation: Multi-branch design increases complexity; lower performance than later methods.

AOD-Net

Li et al., 2017

PSNR 21.01 dB · SSIM 0.837

Approach:
Reformulates the scattering model into a single CNN with a unified parameter $K(x) = (J(x) - A) / t(x) + A$, avoiding separate estimation steps.

⚠ Limitation: Still tied to the physical model; struggles with real-world non-homogeneous haze.

GFN (Gated Fusion)

Ren et al., 2018

PSNR 22.44 dB · SSIM 0.884

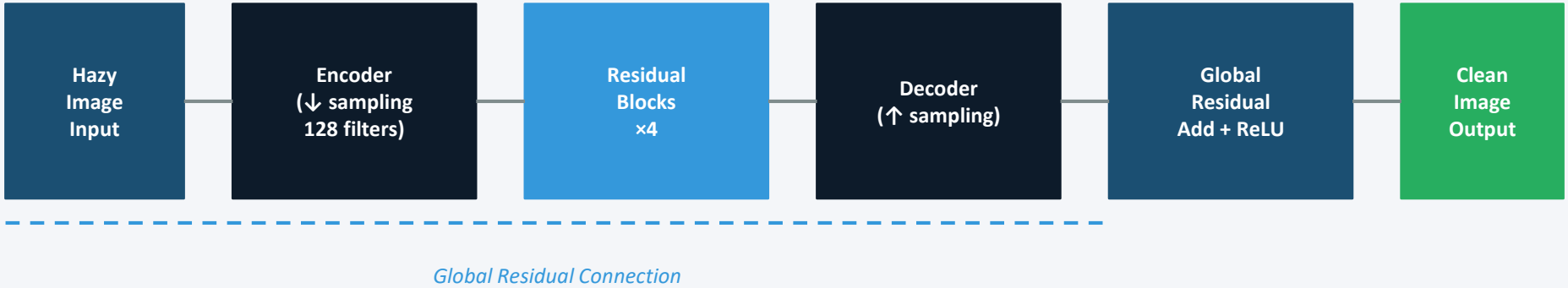
Approach:
Uses three inputs (white-balanced, contrast-enhanced, gamma-corrected) and learns gated fusion weights to combine them into the dehazed output.

⚠ Limitation: Requires careful pre-processing of inputs; fusion weights may not generalise.

→ GMAN addresses these by bypassing the physical model entirely — directly learning I(hazy) → J(clean) end-to-end.

Methodology

Network Architecture – Generic Model-Agnostic Network (GMAN)



Key Design Components:

Encoder: Extracts features, progressively reduces spatial resolution (stride-2 convolutions)
Residual Blocks: Two conv layers + ReLU + skip connection — mitigates vanishing gradient
Decoder: Transposed conv for upsampling, mirrors encoder structure
Global Residual: Adds hazy input to output → network learns only the haze component

Methodology

Loss Function · Dataset · Training Details

Loss Function

Mean Squared Error (MSE):

$L = (1/N) \Sigma \|\hat{J}(x_i) - J(x_i)\|^2$

Measures pixel-wise difference between predicted and ground truth.

Dataset

RESIDE Dataset:

- Indoor Training Set (ITS)
- Synthetic Objective Testing Set (SOTS)

~1000 images used (subset) due to computational constraints.

Training Details

- **Optimizer: Adam**
- Learning Rate: 0.001
- Batch Size: 4
- Input Resolution: 224×224
- Framework: PyTorch
- Epochs: Limited (hardware constraints)

Methodology

Evaluation Metrics

PSNR

Peak Signal-to-Noise Ratio

$$PSNR = 10 \cdot \log_{10}(MAX^2 / MSE)$$

- Measures reconstruction quality
- Higher dB = better quality
- Typical good range: 20–35 dB

Our result: 17.94 dB

SSIM

Structural Similarity Index

$$SSIM(x,y) = l(x,y) \cdot c(x,y) \cdot s(x,y)$$

- Combines luminance, contrast, and structural information
- Range: 0 (worst) to 1 (perfect)
- Reflects perceptual quality

Our result: 0.7133

Results

Quantitative Comparison on SOTS Dataset

Method	PSNR (dB)	SSIM
DCP	18.87	0.794
DehazeNet	22.66	0.833
MSCNN	20.01	0.791
AOD-Net	21.01	0.837
GFN	22.44	0.884
GMAN (Original)	27.94	0.897
Ours (Proposed)	**17.94**	**0.7133**

17.94 dB

Our PSNR

0.7133

Our SSIM

Performance gap attributed to: limited training data (~1000 images), fewer epochs, simplified architecture, and 224×224 resolution constraint.

Results

Qualitative Comparison

Input: Hazy Image

Degraded visibility, low contrast

Output: Dehazed (Ours)

Improved clarity and contrast

Ground Truth: Clear Image

Target reference image

Conclusion

Summary

✓ Implemented a simplified GMAN for end-to-end dehazing

✓ Trained on RESIDE (ITS subset), evaluated on SOTS

✓ Achieved PSNR 17.94 dB and SSIM 0.7133

✓ Model visually improves clarity despite constrained resources

Limitations

✗ Small training set (~1000 images)
✗ Limited training epochs
✗ Simplified architecture vs. original GMAN

Future Scope

1. Train on full RESIDE dataset for better generalization
2. Incorporate attention mechanisms (CBAM, SE blocks)
3. Add perceptual + adversarial loss components
4. Extend to real-world outdoor haze datasets
5. Multi-scale training for improved resolution handling

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